

Entity-Linkage / Sample-Construction Boundary

***Record Matching, Population Formation, and the
Governance of Analytical Samples***

Version: 0.1

Date: July 2026

Status: Preprint / OSF external-facing draft

Context: Meta-Writing Ecology

Document Type: Protocol-Facing Boundary Note

Huang Tzuyuan

Abstract

This document formalizes the **Entity-Linkage / Sample-Construction Boundary** as a protocol-facing note for research and administrative systems that connect records believed to refer to the same person, organization, event, household, place, or other entity. The central claim is that entity linkage is not only a technical matching step. It is also a sample-construction operation.

Linkage rules determine who enters the analytical population, which records are merged, which identities remain separate, which uncertain matches are discarded, and which histories become attributable to a single entity. False matches, missed matches, identifier instability, data-source coverage, threshold selection, clerical review, and temporal change can therefore alter the composition of the sample and the claims later made from it.

The note distinguishes matching accuracy from sample validity, record identity from entity identity, and linkage completion from population adequacy. It does not provide a specific linkage algorithm or statistical estimator. It establishes a boundary that should remain visible in study design, data documentation, validation, reporting, and downstream interpretation.

The framework is relevant to longitudinal research, health and social data, public administration, bibliometrics, customer records, fraud detection, and other settings in which linked data are treated as if the analytical population pre-existed the linkage procedure. It emphasizes that the linked sample is produced through decisions and uncertainty, not merely discovered.

Keywords

Entity Linkage; Sample Construction; Record Linkage; Population Formation; Matching Error; False Match; Missed Match; Identity Resolution; Analytical Sample; Linkage Threshold; Data Governance; Uncertainty; Longitudinal Data; Meta-Writing Ecology.

1. Introduction

Record linkage is often described as a preprocessing task. Data are cleaned, identifiers are compared, probabilities are calculated, and matching records are joined. Once the linked dataset exists, later analysis may treat it as a neutral representation of the relevant population.

This sequence hides a structural fact: linkage creates the object that subsequent analysis studies. Every threshold, blocking rule, identifier choice, exclusion, and manual decision can alter who is present, how histories are combined, and which cases remain unresolved. The technical procedure therefore participates in population formation.

This boundary note makes that participation explicit. It asks not only whether records were matched accurately, but how matching decisions constructed the sample, distributed uncertainty, and bounded the claims that can follow.

2. Core Definition

Entity-Linkage / Sample-Construction Boundary describes the condition in which records, names, identifiers, labels, entities, roles, or source entries are joined

across datasets, systems, archives, platforms, or evidence layers, while the linkage decisions remain hidden, under-documented, unstable, or treated as neutral preprocessing.

The boundary problem appears when a linked dataset, merged table, resolved entity, or constructed sample is treated as analytically ready before the entity relation has been sufficiently defined, validated, documented, and made auditable.

The issue is not that entity matching occurs. Entity matching may be necessary, legitimate, useful, and methodologically appropriate.

The issue appears when matching choices are treated as technical cleanup rather than sample construction.

In this condition, the visible dataset appears coherent because records have been joined. However, the underlying linkage layer may contain judgment calls about name variation, identifier equivalence, parent–subsidiary relation, role level, temporal validity, duplicate handling, threshold choice, candidate rejection, automated classification, or human override.

The central structural claim is:

evidence is not only analyzed; evidence is assembled.

A dataset does not become evidential only at the modeling, interpretation, or publication stage. It becomes evidential during linkage.

3. Classification and Scope

This document presents **Entity-Linkage / Sample-Construction Boundary** as a protocol-facing boundary note. It identifies the conditions a reporting, review, or

documentation procedure should keep visible, but it is not itself a complete protocol, checklist standard, certification system, or compliance instrument.

The note can support method design and audit preparation without prescribing a single technical implementation. The public version omits non-public editorial records, private source materials, non-public working materials, and implementation details.

4. Central Distinctions

entity linkage $\neq$ entity equivalence

Expanded:

A record can be linked to another record without establishing that both records refer to the same entity, role, level, time-bound identity, or evidential unit.

Second distinction:

data merge $\neq$ neutral preprocessing

Expanded:

A merge operation may alter sample composition, evidential scope, analytic inclusion, and downstream inference.

5. Core Mechanism and Structural Sequence

5.1 Core Mechanism

multiple data sources become useful together $\rightarrow$ shared identifiers are missing, partial, unstable, or mismatched $\rightarrow$ candidate links are generated $\rightarrow$ similarity,

identifier, semantic, or manual rules are applied → records become joined → joined table appears clean → matching choices disappear into preprocessing → sample construction becomes difficult to inspect → downstream analysis inherits hidden linkage decisions

5.2 Structural Sequence

Failure Flow

fragmented sources → entity references normalized → candidate matches generated → linkage accepted by score / rule / judgment → merged sample appears complete → matching discretion is not documented → sample construction becomes opaque → downstream estimate or interpretation appears more stable than its linkage layer

Aligned Flow

fragmented sources → target entity definition declared → allowable equivalence specified → candidate generation rules logged → matching method selected by linkage condition → thresholds and review rules documented → validation set or audit sample constructed → precision / recall / sensitivity reported → bridge table and decision log archived when permissible → merged sample remains inspectable

6. Structural Components

source dataset target dataset entity definition entity level identifier system canonicalization rule candidate generation blocking rule similarity score threshold choice manual adjudication automated classifier human-over-the-loop supervision validation set precision / recall false positive link false negative non-link temporal validity bridge table audit log sample attrition downstream inference

7. Primary Failure Patterns

1. Name-as-Entity Collapse

same or similar name → treated as same entity

A label becomes the entity itself.

2. Identifier Overtrust

identifier match → conceptual equivalence assumed

A shared or mapped identifier is treated as sufficient even when the entity level or time validity differs.

3. Clean Merge False Legibility

merged table appears orderly → linkage uncertainty disappears

The table's cleanliness suppresses attention to the construction layer.

4. Threshold Discretion Disappears

similarity threshold chosen → sample composition changes → threshold choice unreported

A discretionary boundary becomes invisible inside the resulting dataset.

5. Candidate Loss Before Review

candidate generation filters records → downstream classifier never sees missing true matches

Failure occurs before the validation layer can observe it.

6. Parent–Subunit Substitution

subunit / brand / local entity → mapped to parent entity → role difference hidden

A higher-level entity absorbs the evidence role of a lower-level actor.

7. Temporal Identity Flattening

current name or identifier → applied backward across time

Entity continuity is imposed after the fact.

8. Human-Over-The-Loop Without Support

machine generates candidate links → human remains responsible → audit

structure is incomplete

Delegated execution occurs without sufficient retained answerability support.

8. Operational Evaluation

1. What is the target entity?
2. What level is being matched: record, person, organization, product, brand, document, citation, parent entity, subunit, platform profile, or time-bound identity?
3. Are the linked records actually equivalent, or only related?
4. What counts as allowable equivalence?
5. Is the match one-to-one, one-to-many, many-to-one, or many-to-many?
6. Are temporal validity rules specified?
7. Are name changes, aliases, parent–subsidiary relations, ownership changes, or role shifts handled explicitly?
8. What candidate generation rules were used?
9. What records were excluded before matching?
10. What threshold, score, or review rule accepted the match?
11. What validation set or audit sample supports the match quality?
12. Are false positives and false negatives reported?

13. Are unmatched records retained, routed, or silently dropped?
 14. Can another reviewer reconstruct the matching decision?
 15. Does downstream analysis depend on the matched sample as if the linkage layer were stable?
-

9. Application Domains

1. Longitudinal research

Linkage across time can create or break individual histories, affecting attrition, exposure, and outcome measurement.

2. Public and administrative data

Identity systems and record integration can determine eligibility, service histories, and institutional visibility.

3. Health and social research

Uneven identifiers and source coverage can produce differential linkage quality across groups.

4. Bibliometrics and organizational analysis

Author, institution, and publication identities can be merged or split in ways that reshape networks and productivity claims.

5. Fraud, compliance, and risk systems

Aggressive linkage can concentrate suspicion on merged identities, while conservative linkage can miss distributed patterns.

10. Relations to Adjacent Public Frameworks

1. Provenance–Validity Separation Model

Provenance–Validity Separation Model separates traceable origin from the validity, adequacy, or warranted use of an output. The present entry extends that distinction into a more specific interface, research, or responsibility condition.

2. Responsibility Alignment Model

Responsibility Alignment Model clarifies whether ownership, authority, capacity, visibility, and accountability remain attached to compatible nodes. The present entry specifies one condition under which that alignment can fail or become nominal.

3. Boundary Failure Diagnostics

Boundary Failure Diagnostics provides a general vocabulary for locating non-entry, closure, lock-in, and boundary loss. The present entry narrows that diagnostic field to a repeatable mechanism.

4. Cost Visibility and Redistribution Model

Cost Visibility and Redistribution Model tracks how costs become visible at one node and hidden or displaced at another. The present entry shows how verification, correction, access, or responsibility costs can be redistributed by the relevant interface.

5. Observer–Representation Boundary

Observer–Representation Boundary separates what an observer can read from what a representation can legitimately establish. The present entry specifies the evidential or operational consequences of crossing that boundary.

11. Boundary Conditions and Non-Applicability

11.1 Boundary Conditions

This note applies when entity matching, record linkage, data merging, identity resolution, metadata reconciliation, or cross-source alignment affects what enters the analytic sample.

The boundary is crossed when:

linked records → treated as stable entities

or:

merged dataset → treated as neutral evidence base

without sufficient disclosure of the matching logic.

It does not apply merely because two datasets are joined. It applies when linkage decisions materially shape evidence while remaining under-specified or unauditable.

11.2 Non-Applicability

This note should not be used to claim that all record linkage is invalid, all automated matching is unreliable, or all merged datasets are suspect.

It does not apply well when:

shared stable identifiers exist

entity definitions are clear

equivalence rules are explicit

matching does not affect analytic inclusion

linkage uncertainty is low and disclosed

validation is proportionate to consequence

unmatched cases are logged rather than silently dropped

the merged data is used only for low-consequence exploration

The narrower concern is:

sample construction becomes hidden when linkage decisions are treated as neutral preprocessing.

Relation to Protocol Function

This Note names the boundary.

The protocol-facing function specifies the minimum audit layer required when the boundary is active.

Entity-Linkage / Sample-Construction Boundary Note → identifies the structural risk

Entity Matching Auditability Protocol Anchor → specifies the reporting and archival function

Compact relation:

boundary note

what can go wrong

protocol anchor

what must remain inspectable

12. Public Orientation and Method Boundary

This document is designed as a public conceptual entry. It provides a definition, mechanism, diagnostic orientation, and limit statement. It omits non-public editorial

records, private source materials, non-public working materials, and implementation details.

The entry is not a substitute for empirical research, legal analysis, professional judgment, technical testing, or domain-specific governance. It should not be used to score people or institutions, automate high-stakes decisions, or infer misconduct from conceptual similarity.

Examples and application domains are illustrative. A case should be classified under **Entity-Linkage / Sample-Construction Boundary** only when the relevant structure can be demonstrated from available evidence.

13. Limitations

1. No algorithmic prescription

The note does not select deterministic, probabilistic, privacy-preserving, or machine-learning linkage methods.

2. No universal error threshold

Acceptable false-match and missed-match rates depend on consequence and use.

3. Ground truth can be unavailable

Validation data may itself be incomplete or institutionally produced.

4. Linkage is only one source of sample bias

Coverage, measurement, consent, missingness, and attrition also shape the sample.

5. Legal and ethical requirements vary

The framework does not replace domain-specific privacy, consent, or data-governance rules.

14. Minimal Formulation

entity name $\neq$ entity identity

record linkage $\neq$ construct alignment

candidate match $\neq$ validated match

matched table $\neq$ transparent sample construction

identifier availability $\neq$ conceptual equivalence

high similarity score $\neq$ correct linkage

automated matching $\neq$ audit completion

human review $\neq$ unrestricted responsibility transfer

clean merge output $\neq$ completed verification

matching choice $\neq$ neutral preprocessing

parent entity $\neq$ local actor by default

temporal alignment $\neq$ entity continuity

same label $\neq$ same role

linked evidence requires linkage evidence

a dataset is not only analyzed; it is assembled

Matching is not preprocessing; it is sample construction.

Alternative compressed form:

Evidence changes when the linkage layer is hidden.

15. Conclusion

Entity linkage is simultaneously a matching operation and a sample-construction operation. Treating it only as technical preprocessing hides the decisions that determine population membership, historical continuity, and uncertainty.

A structurally adequate account reports how entities were defined, how matches were made and reviewed, how uncertainty was handled, and how linkage error could affect the claims made from the resulting sample.

Naming Declaration

Entity-Linkage / Sample-Construction Boundary is a Meta-Writing Ecology term for the boundary between linking records and constructing an evidential sample.

The term entity-linkage refers to the process by which records, names, identifiers, labels, citations, profiles, products, organizations, documents, people, or other units are treated as referring to the same or related entity across source systems.

The term sample-construction boundary refers to the point at which linkage decisions begin to determine what enters the analytic object, what is excluded, what is treated as equivalent, and what downstream claims can be made.

This term does not refer to any specific discipline, dataset type, matching technology, market, institution, provider, software system, or empirical domain.

Within Meta-Writing Ecology, the term functions as a protocol-facing note. It clarifies how data construction, verification labor, responsibility alignment, false legibility, proxy substitution, and AI-readable auditability intersect when linked records become evidence.

It should be used when a merged dataset appears ready for analysis but its linkage layer remains insufficiently visible, validated, or auditable.

Entity Matching Auditability Protocol Anchor